Multiphysics Simulation of Human Deep Venous Valve Mechanics and Deep Vein Thrombosis Development

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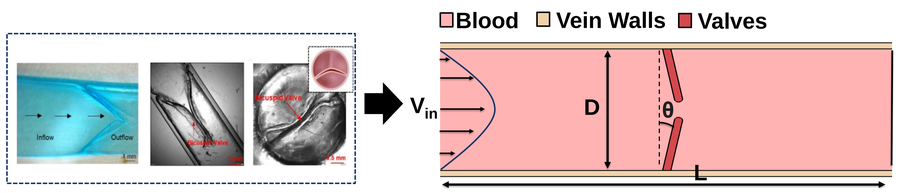
Objectives: (1) Develop a **multiphysics model** in COMSOL with fully coupled Fluid Structure Interaction (FSI) to predict and model vein hyperelasticity, blood clot interaction, and dynamic flow patterns. (2) Perform parametric analysis on valve and clot parameters to assess conditions influencing susceptibility to **Deep Vein Thrombosis (DVT)** blood and to inform DVT treatment approaches. (3) Utilize FEA simulation for comprehensive **mechanics** and **hemodynamic** analysis.

Introduction and Motivation

Vascular diseases, such as atherosclerosis, aneurysms, peripheral artery disease (PAD), and thrombosis, account for nearly 17 million annual deaths around the world. Deep Vein Thrombosis (DVT) in particular, is an especially preventable cause of death that is estimated to affect upwards of 1 million per year and killing nearly 100,000 per year in the US alone. DVT is a condition where blood clots form in the deep slow-moving especially in the legs. Regions of slow moving fluid and low vorticity commonly in the sinus region of the valves is where clots are described to commonly form. Fluid Structure Interaction (FSI) and Multiphysics analysis can provide effective ways to better understand conditions that can increase progression of DVT and cause clot detachment by prediction hemodynamics and valve/clot mechanics



Current Literature Simulation and Limitations

- Valves are often treated **as linear elastic** and valve walls as static boundaries
- Recirulation zones are commonly analyzed through Langrangian methods to track particles
- Valve and vein walls can undergo **large deformation** especially in very small deep veins that cannot be accurately modeled through linear elastic analysis

Methodology and Modelling

Model Assumptions

- Flow is **incompressible**
- Valves and Wall are flexible, and a **hyper-elastic material** model is suitable
- **Laminar** flow and **fully developed** before valve
- Reynolds Number (Re) = $O(10^1)$
- Flow is **fully developed** before reaching the valves
- Blood is treated as **Newtonian**

Region	C10 (kPa)	C01 (kPa)	K (Pa)	ρ (kg/m³)
Valve	50	6	1.3e6	1100
Wall	83.137	1764.5	1.2e6	1200

Fluid	ρ (kg/m³)	η (Pa·s)
Blood	1060	0.0035

Table 1. Mooney Rivlin material constants for hyper-elastic materials and blood properties

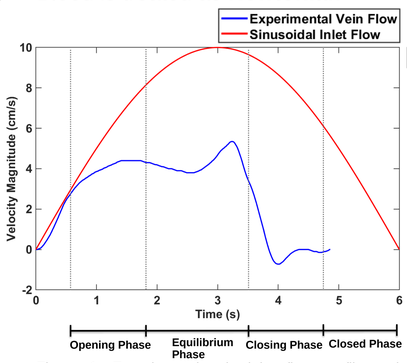


Figure 1. Experimental vein inlet flow profile and simplified sinusoidal fit for pulsatile flow.

COMSOL Multiplysics 6.2 was used to solve the transient Navier stokes equation. A time varying velocity inlet and a 0 pressure outlet boundary conditions was applied for all simulations. Two way coupling Fluid Structure Interaction (FSI) predicted fluid structure force and a deformable domain applied to conform the fluid domain to the deforming solid domain. All simulation was 2D with triangle elements for all structures. Boundary layers elements and refinement was around valve structure and where no slips conditions exist. Material properties and inlet velocity profiles were taken from existing experimental stress strain testing and literature.

Results – Valve Angle Parameter Study

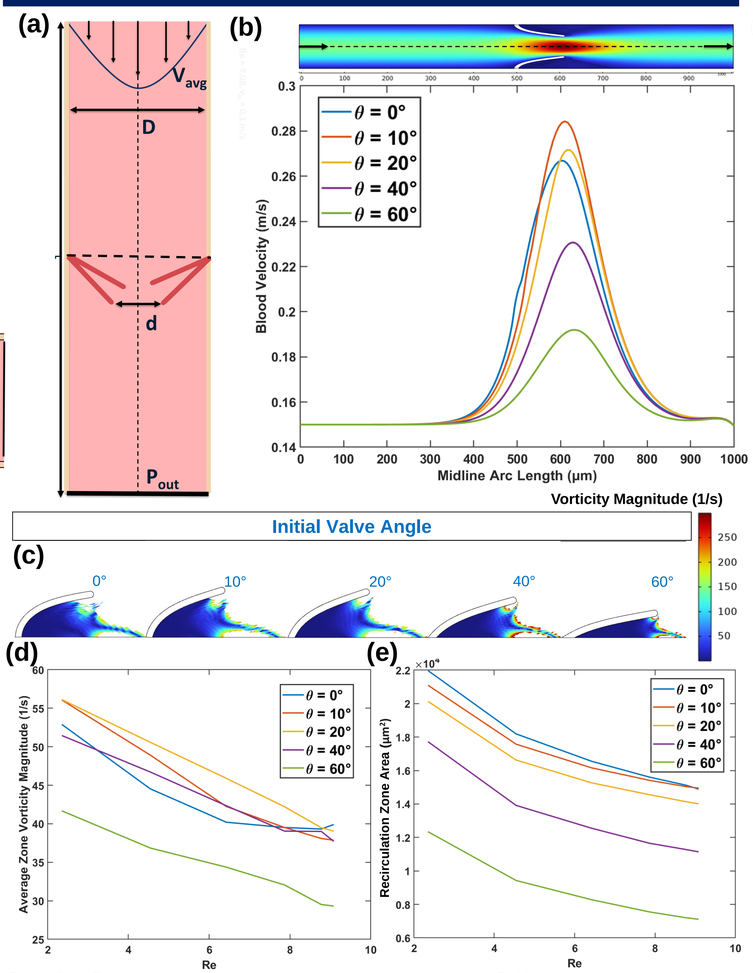


Figure 2. A) Parameter study initial valve angle and parameter variables. B) Midline blood velocity magnitude at Re = 9.81. C) Recirculation zone visualization for different initial valve angles (Re = 9.81). D) Average recirculation zone vorticity magnitude vs Re. E) Average recirculation zone area vs Re.

Results - Clot Parameter Study

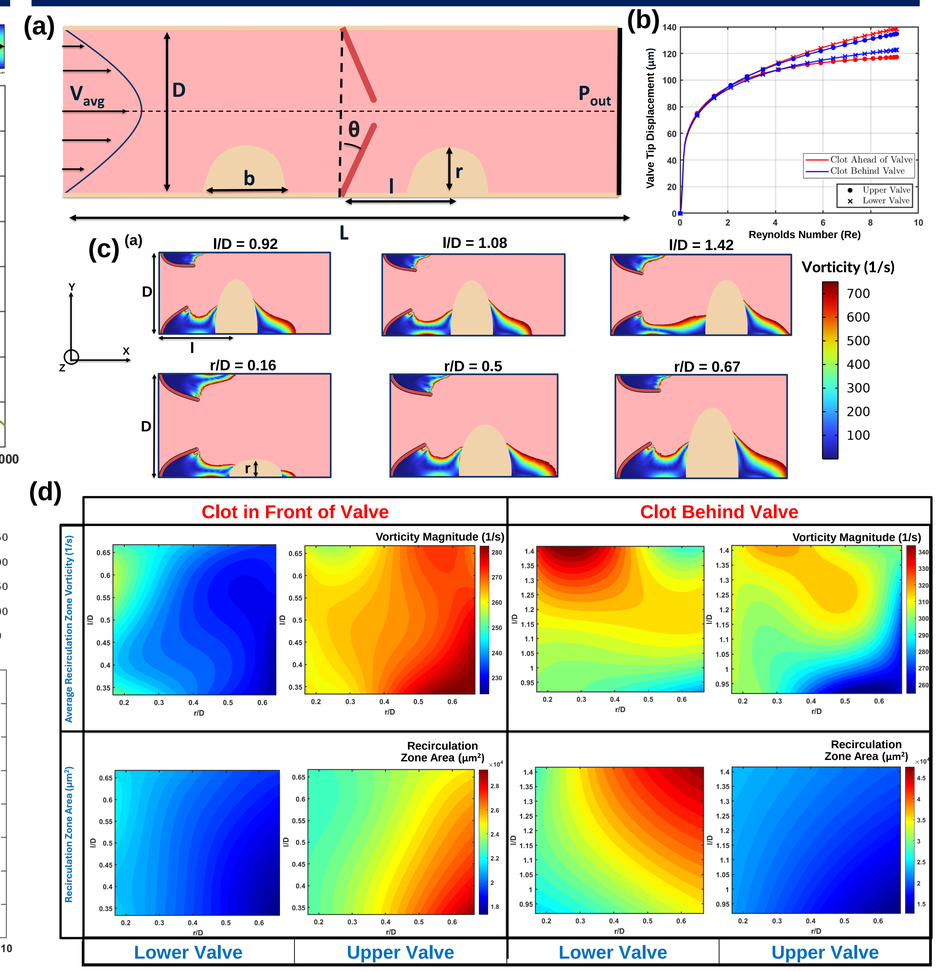


Figure 3. A) Parameter study clot distance, size, and location. B) Valve Tip deflection vs clot location (Re = 9.81) C) Recirulation zone visualization for clot behind valve changing clot size and distance. D) Heat maps of recirculation zone area and vorticity magnitudes for lower and upper vales and for the clot Infront and behind valve case.

Results – Deformable Valve Walls

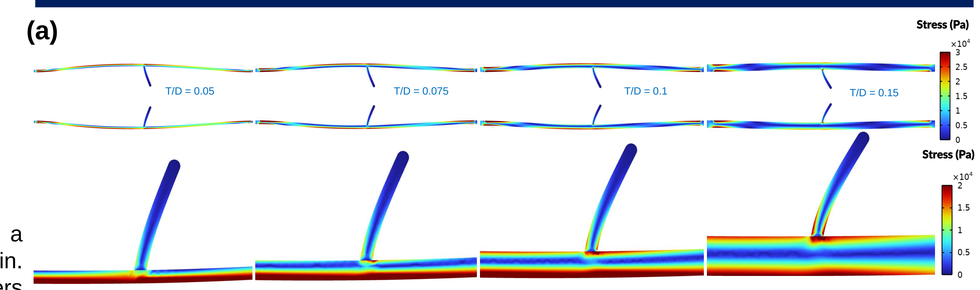


Figure 4. A) Parameter study with deformable vein walls. The top depicts the stress distribution in the vein walls, while the bottom shows the valve stress distribution in Pascals.

Conclusions

- An initial valve angle of **~20° has the best combination** of valve performance, reduced recirculation zone area, and vorticity magnitude.
- Clots in front of valve pose lower average vorticity but smaller recirculation zone areas while clots behind valve have larger average vorticity but larger recirculation zone areas.
- Clot formation has a **greater dependence on clot size** than distance from valve.
- Including deforming walls **reduces valve deflection** predictions and **increases valve wall stresses.**

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